

X-Gen: Enhancing Radiology Report Generation via LLM-Driven Data Augmentation and Decoupled Training

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Abstract

The scarcity and limited accessibility of medical data significantly challenge deep learning applications in medical AI. Radiology report generation (RRG), a key medical AI research area, could greatly improve computer-aided diagnosis through automated X-ray image interpretation. However, obtaining paired X-ray images and reports is labor-intensive and restricted by strict regulations. Large language models (LLMs), such as GPT-4, provide a promising alternative by enabling cost-effective text data augmentation and report rewriting in varied styles. We rigorously assess augmented data’s clinical accuracy and stylistic similarity to radiologist-authored reports through expert evaluations. Interestingly, augmented data enhances RRG model performance, yet performance declines when augmented data surpasses original data volume due to style distribution shifts. To mitigate this, we propose integrating a conditional variational autoencoder (cVAE) into the RRG model to separate medical semantics from writing styles during training, enabling better handling of augmented data’s distribution shift. Our proposed method, X-Gen, combines data augmentation with decoupled training. Tested on two public Chest X-ray datasets and a private abdomen X-ray dataset, X-Gen significantly improves the performance of baseline models, showcasing its effectiveness and versatility in X-ray report generation.

1. Introduction

Automated radiology report generation (RRG) significantly enhances AI-aided diagnostic efficiency, allowing radiologists to focus more on core diagnostic tasks. Rudolph et al. [23] showed that AI-assisted interpretation of chest X-rays matched radiologist-level performance. However, the scarcity of training data, caused by the labor-intensive and

costly collection of annotated radiograph-report pairs, limits its practical deployment.

Previous RRG research mainly improves model architectures, image-text alignment, or incorporates domain-specific knowledge. Data augmentation for generating high-quality reports remains under-explored due to challenges in maintaining narrative coherence, fluency, and clinical accuracy. Large language models (LLMs), such as GPT-4, demonstrate promising capabilities for medical text augmentation. GPT-4’s strong performance on medical benchmarks and successful application in NLP tasks suggest potential suitability for radiology report augmentation. Yet, questions remain regarding the quality and clinical accuracy of LLM-generated reports and the impact of augmented data on model performance.

To address these challenges, we propose X-Gen, a two-step paradigm combining data augmentation and decoupled training. We generate augmented reports by rewriting original reports in different styles using GPT-4, validated through rigorous expert evaluations for coherence and accuracy. While augmented data initially improves model performance, excessive synthetic data causes performance decline due to style distribution shifts. To mitigate distributional shifts, we integrate a conditional variational autoencoder (cVAE) into the generation process, decoupling medical semantics from writing styles. The cVAE quantizes style embeddings through a learned codebook and reconstructs reports non-autoregressively, effectively balancing semantic enrichment and style variation. Applying X-Gen to baseline models R2Gen [4] and CvT2GPT2 [21], we observe significant performance improvements across traditional NLG and advanced clinical efficacy (CE) metrics. For example, X-Gen improves R2Gen’s BLEU-4 score on the IU dataset from 0.165 to 0.206, achieving results comparable to state-of-the-art models. Experiments on additional chest and abdomen X-ray datasets further demonstrate X-Gen’s versatility and effectiveness.

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2. Related Work

2.1. Data Augmentation

Data augmentation techniques tackle the issue of data scarcity and limited data diversity by synthesizing training data. It is a cost-effective alternative to data collection, effectively enhancing model performance. Confronted with the inherent intricacies of natural language, numerous sophisticated techniques, including [20, 24, 37] have been developed.

All these methods operate at the sentence level and thus are not inherently suited for radiology report augmentation scenarios. For instance, a radiology report tends to be a long narrative *i.e.* a paragraph rather than a sentence, that conveys a substantial amount of medical terminologies with uncertainty. Word- or sentence-level perturbation is at the risk of altering the original meaning of the report.

Most recently, given well-designed prompts, AugGPT [5] exemplifies that ChatGPT can surpass existing data augmentation techniques on multiple NLP benchmarks. The data synthesized by ChatGPT attains elevated scores in embedding similarity and data learnability. Consequently, it is reasonable to presume that GPT-4 could yield superior data augmentation outcomes compared to ChatGPT.

2.2. Radiology Report Generation

In general vision-language tasks, image captioning is considered the most similar to radiology report generation. Both tasks involve function modules such as image understanding, image-text alignment, and text generation *etc.* Image captioning seeks to use short, coherent, and diversified natural language to describe salient objects and their attributes in RGB images. In contrast, authoring a radiology report goes beyond common sense, requiring domain knowledge and years of professional experience. To address the domain-specific challenges, researchers have developed numerous techniques. Many previous works have focused on engineering sophisticated model architectures, drawing inspiration from image caption models.

Some earlier notable works, such as [35, 13], all employ LSTM [9] as the generation decoder. Later, since transformers have become ubiquitous in model design for various tasks, radiology report generation (RRG) models have begun replacing RNNs with transformer blocks as decoders. R2Gen [4] adopts a transformer-based encoder-decoder architecture with a customized memory module designed to capture language patterns and update them step-by-step.

3. Method

As shown in Figure 1, X-Gen involves LLM-driven data augmentation and a dual-branch architecture: a cVAE branch which consists of a style encoder $\mathcal{E}_s$ and a style decoder $\mathcal{D}_s$, and the generation branch contains an image en-

coder $\mathcal{E}_g$ and a report decoder $\mathcal{D}_g$. Two branches are connected via a codebook Ω , *i.e.*, a discrete latent space, which contains learned style embeddings from both original data and augmented ones.

Each report is paired with two radiographs in the IU and abdomen X-ray datasets. We extract the image features and concatenate them together as x . The number of radiographs associated with each report may vary for the MIMIC-CXR dataset. The common practice is treating each radiograph with the same report as a distinctive sample. During MIMIC-mini construction, we ensure each report only appears once. We also denote the single image feature as x .

During training, the style encoder in the cVAE branch takes all reports paired with x , both original and augmented, as inputs. We randomly sample one report for the generation branch as a training target. During inference, the cVAE branch is discarded. The decoding process of the generation branch works as usual, except for selecting a style embedding from the codebook as a control signal. The detailed process is outlined as follows.

3.1. Data Augmentation with GPT-4

Prior research [34] has demonstrated that the behavior of LLMs is sensitive to prompt design. Our initial experiments reveal a similar phenomenon. We begin by instructing GPT-4 to rewrite radiology reports while maintaining the medical terminologies intact. Upon reviewing its outputs, we notice that GPT-4 sometimes adds style tags such as “*causal*”, “*informal*”, or “*conversational*” at the beginning of the report. These tags are accompanied by phrases like “*Chest X-ray looks good*”, “*nothing to worry about*”. Additionally, some outputs labeled with the “*lengthy*” tag include redundant content such as “*upon careful examination of the given chest X-ray*”.

This observation led us to specify rewriting styles to improve the quality of synthetic data. Intuitively, we opted for style tags such as “*formal*” and “*professional*” to exclude colloquial expressions from completions and “*concise*” and “*detailed*” to act as length control signals. However, we also want to ensure that the styles of synthetic data remain aligned with actual medical practice. To achieve this, we randomly sampled 20 reports from the IU X-ray training set and task GPT-4 to rewrite each report in the following styles: “*comprehensive and succinct*”, “*comprehensive and detailed*”, “*professional and concise*”, “*formal and concise*” and “*formal and detailed*”. A radiologist then reviewed these styles and selected the most appropriate tags based on his expertise and preference. In Section 1 of the supplementary materials, we provide examples of augmented reports from the IU training set.

“*professional and concise*”, “*formal and concise*” and “*formal and detailed*” are chosen for large-scale experi-

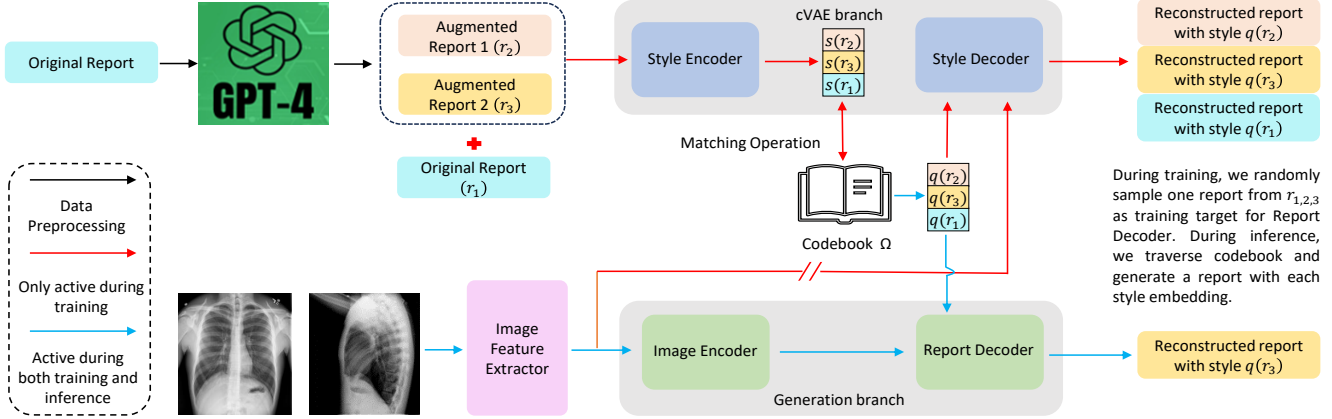


Figure 1. An overview of our proposed method. We extract the [STY] token from the output of the style encoder. The matched entries from the codebook are treated as style embeddings of each report. Style embeddings and image features are the inputs of the style decoder, which are used together to reconstruct reports. There is no gradient flow between the style decoder and the image feature extractor.

ments. We adopted OpenAI’s official ChatCompletion template, and the final version of our prompt is shown in Figure 2.

“role”: “system”, “content”: “The following is a chest X-Ray report. Rewrite the report in different styles while keeping the medical terminologies intact. The styles are 1. professional and concise, 2. formal and concise. 3. formal and detailed”, “role”: “user”, “content”: “[original report]”

Figure 2. The prompt used for rewriting reports in different styles.

3.2. Decoupled Training

Conditional Variational Autoencoder (cVAE) [26] was introduced to generate data conditioned on auxiliary information, such as generating MNIST digits conditioned on class labels. To be more specific, the encoder of cVAE first compresses the input data into the latent space, a lower-dimensional and structured representation of the input data. Its decoder then takes the latent variable sampled from the latent space and the condition to generate or reconstruct data that matches the condition.

In our case, cVAE is well suited as an unsupervised approach for decoupling the medical semantics from the writing styles, which includes the ones we specified in the prompt for GPT-4 and the unknown amount of styles in the training data. Similar to the [CLS] token used in BERT [7], we append a learnable [STY] token at the beginning of each report as the neural representation of its style, which is compressed into the latent space through the encoder of the cVAE branch. To ensure the learning of distinct styles, the decoder of the cVAE branch is tasked to reconstruct each report based on the sampled latent variable conditioned on the X-ray image paired with the report.

Another non-trivial aspect of the decoupled training is the design of latent space. There are two design choices for the latent space: continuous and discrete. A continuous latent space is the distributional representation of all writing styles in the combined data set, making distinguishing styles of original data from synthetic data challenging. On the other hand, a discrete latent space, i.e., a codebook, allows for the individual selection of specific writing style. This feature is especially desirable during inference, as we want the report generated by our model to closely align with ones written by radiologists.

In the following paragraphs, we explain the details of our decoupled training.

3.2.1 Style Encoding and Assignment

We define x as the combined image features, r_i in $\{r_i\}_{i=1}^n$ as the corresponding report, Ω as codebook. During training, inputs of the encoder $\mathcal{E}_s$ of the cVAE branch are x and r_i prepended with a special token [STY]. The hidden state of the [STY] token from the last layer of $\mathcal{E}_s$ is treated as the style representation of r_i , denoted by $s(r_i)$. Then we find the closest match in Ω of $s(r_i)$, denoted by z_i . The training objective is defined as:

$$\max_{\mathcal{E}_s, \mathcal{D}_s, \Omega} \log p(r_i | z_i, x), \text{ where } z_i = \text{Match}(s(r_i), \Omega) \quad (1)$$

Here, $p(r_i | z_i, x)$ is the conditional distribution of the report r_i given its style embedding z_i and the paired image features x , which is approximated with the style decoder $\mathcal{D}_s$. $\text{Match}(\cdot, \cdot)$ indicates the matching operation. As no gradient is defined for matching operators, the gradients from z_i are directly copied to $s(r_i)$ with the same technique used by [27]. For the matching operation, we adopt a similar approach proposed in [2] for object query assignment. For each x and paired reports $\{r_i\}_{i=1}^n$, we construct a bipartite graph between $\{s(r_i)\}_{i=1}^n$ and the style embeddings in Ω .

The goal is to find a permutation of k elements $\tau \in \mathfrak{S}_k$ with the lowest assignment cost:

$$\hat{\tau} = \arg \min_{\tau \in \mathfrak{S}_k} \sum_i^k \|s(r_i) - \Omega_{\tau(i)}\|_2 \quad (2)$$

and the assigned style embedding for each r_i is $z_i = \Omega_{\hat{\tau}(i)}$. Subsequently, the Vector Quantization algorithm is employed to optimize the alignment of z_i with $s(r_i)$, utilizing the mean square loss as a metric to train the associated codebook entries. The final objective function is formed as:

$$-\log p(r_i|z_i, x) + \|\text{sg}[s(r_i)] - z_i\|_2^2 + \beta \|s(r_i) - \text{sg}[z_i]\|_2^2 \quad (3)$$

where the first term inherited from Eq. (1) is used to optimize $\mathcal{E}_s$ and $\mathcal{D}_s$. The second term is for training the matched entries in Ω . The final term is a commitment loss, which constrains the embedding space of Ω . $\text{sg}[\cdot]$ refers to “stop gradient” and β is a hyper-parameter.

3.2.2 Fully Non-autoregressive Decoding

Style embeddings are trained through a fully non-autoregressive (NAT) objective, as the decoder should be able to reconstruct reports based solely on the image features and style embeddings. In an auto-regressive setting, the objective function is usually formulated as:

$$-\log p(r_i|z_i, x) = \sum_{t=1}^T -\log p(w_t|w_{j<t}, z_i, x) \quad (4)$$

at step t , when predicting token w_t , the decoder has access to all preceding ground-truth tokens $w_{j<t}$, which can be considered as a style information leakage. As a result, the training signals for the style embeddings could be futile. Instead, the style decoder $\mathcal{D}_s$ takes z_i, x , and a sequence of T [MASK] tokens as inputs, and predicts each target token in r_i in a conditionally-independent manner, *i.e.*,

$$-\log p(r_i|z_i, x) = \sum_{t=1}^T -\log p(w_t | [\text{MASK}], z_i, x) \quad (5)$$

and estimate the conditional distribution $p(r_i|z_i, x)$ in Eq. (3).

3.3. Generation Branch

The cVAE branch is an agile design that can be integrated into most existing report generation models as a cost-free plugin. To showcase the effectiveness of our method, we simply choose two generation models, *e.g.*, R2Gen [4] and CvT2GPT2 [21] as our baseline models. Adding the cVAE branch to the baseline model only requires a minor modification to the token embedding layer of $\mathcal{D}_g$ by adding the style embedding z_i to the word embeddings of each token in r_i . The training of two branches is performed jointly, with the objective functions in Eq. (3) and Eq. (4), respectively. To make the convergence speed of the two branches compatible, for each x , the cVAE branch takes all its paired

reports $\{r_i\}_{i=1}^n$, while the generation branch randomly samples just one report from $\{r_i\}_{i=1}^n$.

4. Experiments

4.1. Experiment Settings

4.1.1 Datasets

IU Chest X-ray Indiana University chest X-ray collection [6] is the most widely used X-ray report generation benchmark, containing 3,955 radiology reports and 7,470 chest X-ray radiographs. Strictly following the setup and dataset partition of previous works [4, 32], we end up with 2,069/296/590 samples for training/validation/test, respectively. The average length of reports in IU X-ray is 222 characters.

Abdomen X-ray The abdomen dataset¹ is constructed in the same format as the IU X-ray dataset with 2,048/500/500 samples for training/validation/test, respectively. Each sample contains two radiographs covering different abdomen areas and a radiology report. The average length of the abdomen report is 217 characters. All images have been resized to 224×224 .

MIMIC-mini MIMIC-CXR [14] contains 377,110 JPG format images and 227,827 free-text radiology reports. Its large scale is out of the scope of our setting as we aim to explore a data augmentation technique where there is not enough training data. Hence, we constructed MIMIC-mini, which contains 5,000 training samples, 296 validation samples, and 590 test samples. We strictly follow the official data partition of MIMIC-CXR to ensure that our training, validation, and test samples are sourced exclusively from the original sets. Each report only appears once in MIMIC-mini and is associated with one image. The average length of reports in MIMIC-mini is 424 characters.

4.1.2 Evaluation Metrics

For NLG metrics, we use BLEU [22], CIDEr [29], ROUGE-L [17], and METEOR [1]. For CE metric, we employ Bert-Score (Bert-S) [38], CheXbert micro F1 for all observations (F1) [25], and Radgraph-related metrics, *i.e.*, RadEntity-MatchExact (RE-ME), RadEntity-NLI (RE-NLI) [19] and Radgraph-complete (Rad-C) [12].

4.1.3 Implementation Details

For data augmentation, to comply with PhysioNet’s guide of “Responsible use of MIMIC data with online services like GPT”, we used Azure OpenAI service and opted

¹Ethics approval for private data used in this study was provided by Anonymous Human Research Ethics Committee.

Table 1. Main results on IU X-ray, MIMIC-mini, and Abdomen X-ray. By the time of the experiment, KiUT[11] and METransformer[32] have not open-sourced their code. We put the best result in bold and underlined the second-best result.

Dataset	Methods	YEAR	B-1	B-2	B-3	B-4	M	R-L	C
IU-Xray	R2Gen[4]	2020	0.470	0.304	0.219	0.165	0.187	0.371	-
	R2GenCMN [3]	2022	0.470	0.304	0.222	0.170	0.191	0.358	-
	MSAT[33]	2022	0.481	0.316	0.226	0.171	0.190	0.372	0.394
	XPRONET[30]	2022	0.525	0.357	0.262	0.199	0.220	0.411	0.359
	COMG[8]	2023	0.473	0.305	0.217	0.162	0.186	0.378	-
	KiUT[11]	2023	0.525	0.360	0.251	0.185	0.242	0.409	-
	R2GenGPT[31]	2023	0.488	0.316	0.228	0.173	0.211	0.377	0.438
	METransformer[32]	2023	0.483	0.322	0.228	0.172	0.192	0.380	0.435
	CvT2GPT2[21]	2023	0.473	0.303	0.224	0.175	0.199	0.376	0.694
	X-R2Gen-1	Ours	0.484	0.338	0.246	0.184	0.213	0.409	0.472
	X-R2Gen-2	Ours	<u>0.522</u>	<u>0.369</u>	<u>0.272</u>	<u>0.206</u>	0.218	0.406	0.488
	X-CvT2GPT2-1	Ours	0.497	0.341	0.252	0.195	0.211	0.401	0.588
	X-CvT2GPT2-2	Ours	0.525	0.370	0.275	0.209	<u>0.240</u>	<u>0.409</u>	<u>0.655</u>
MIMIC-mini	R2Gen[4]	2020	0.366	0.224	0.149	0.104	0.139	0.281	0.143
	R2GenCMN[3]	2022	0.361	0.213	0.137	0.093	0.141	0.270	0.117
	XPRONET[30]	2022	0.333	0.204	0.131	0.095	0.127	0.265	0.112
	COMG[8]	2023	0.358	0.216	0.144	0.103	0.134	0.271	0.126
	M2KT[36]	2023	0.357	0.228	0.151	0.108	0.137	0.278	0.141
	ORGAN[10]	2023	0.385	0.230	0.157	0.108	0.142	0.283	0.183
	R2GenGPT[31]	2023	<u>0.395</u>	0.242	<u>0.167</u>	0.119	<u>0.155</u>	0.295	0.226
	CvT2GPT2[21]	2023	0.388	0.237	0.161	0.110	0.148	0.277	0.183
	X-R2Gen-1	Ours	0.376	0.231	0.153	0.107	0.142	0.282	0.153
	X-R2Gen-2	Ours	0.380	0.233	0.155	0.109	0.143	0.283	0.156
	X-CvT2GPT2-1	Ours	0.392	0.240	0.164	0.115	0.151	0.282	0.187
	X-CvT2GPT2-2	Ours	0.396	0.242	0.168	0.119	0.156	<u>0.291</u>	<u>0.222</u>
	Transformer[28]	2017	0.408	0.247	0.161	0.107	0.142	0.314	0.261
Abdomen-Xray	R2Gen[4]	2020	0.371	0.234	0.159	0.110	0.142	0.316	0.286
	R2GenCMN[3]	2022	0.361	0.230	0.150	0.102	0.135	0.309	0.161
	MSAT[33]	2022	0.410	0.246	0.156	0.105	0.139	0.286	0.275
	R2GenGPT[31]	2023	<u>0.450</u>	0.282	<u>0.187</u>	<u>0.127</u>	<u>0.162</u>	<u>0.319</u>	0.314
	CvT2GPT2[21]	2023	0.372	0.209	0.123	0.108	0.142	0.264	0.268
	X-R2Gen-1	Ours	0.395	0.256	0.161	0.112	0.148	0.307	0.286
	X-R2Gen-2	Ours	0.439	0.273	0.180	0.122	0.160	0.316	0.292
	X-CvT2GPT2-1	Ours	0.410	0.258	0.170	0.119	0.155	0.311	0.292
	X-CvT2GPT2-2	Ours	0.451	0.282	0.189	0.129	0.167	0.323	<u>0.301</u>

out of human review of the data. To proceed, we employ OpenAI official ChatCompletion API, with the underlying model GPT-4. The temperature is set to 0.2 for minor variations, while both the presence penalty and frequency penalty are set to zero. Data augmentation is **ONLY** conducted within the training split of each dataset *i.e.* IU X-ray, MIMIC-mini and Abdomen X-ray.

For the cVAE branch, we employ the BERT Encoder architecture [7] without pre-trained weights for both style encoder and decoder, with a hidden dimension of 768. The codebook is initialized as an embedding matrix with 32 entries. We use the official code base of R2Gen and CvT2GPT2 for the generation branch.

The image feature extractor used in R2Gen is ResNet-101 with ImageNet-1k pre-trained weight, while CvT2GPT2 uses a Convolutional vision Transformer (CvT) with ImageNet-21K pre-trained weight. For X-R2Gen, we use Adam [16] optimizer with a weight decay of $5e-5$ and

set the learning rate for the visual feature extractor as $5e-5$ and $5e-4$ for the remaining parameters. For X-CvT2GPT2, we use AdamW [18] optimizer with a weight decay of $1e-4$ and set the learning rate for the encoder and decoder as $5e-5$ and $5e-4$, respectively. The majority of our experiments are conducted using a single RTX-3090 GPU.

4.2. Experiment Results and Analysis

4.2.1 Experts Evaluation on Augmented Data

We randomly selected 886 reports from the IU X-ray training set and rewrote each in three styles, generating 2,658 synthetic reports. Each report was then post-processed to remove style tags and control characters.

We collaborate with seven licensed radiologists for their expertise in the evaluation process, which comprises two assessment tasks. The first task evaluates the semantic and clinical accuracy of the augmented reports. In this phase, the original and augmented reports are clearly labeled to

Table 2. Task 2 statistics: The “original” still holds advantages over synthetic styles. “*formal and detailed*” style is the least favored option.

Report Types	Ranking Distributions			
	1st	2nd	3rd	4th
Original	0.412	0.291	0.167	0.128
professional and concise	0.301	0.364	0.182	0.151
formal and concise	0.147	0.180	0.428	0.245
formal and detailed	0.139	0.165	0.221	0.474

distinguish them from each other. The second task involves anonymizing the reports, *e.g.*, we shuffle and label them as reports 1, 2, 3, and 4, making the original report indistinguishable. Reviewers are instructed to rank these reports based on writing quality and consistency, ensuring an unbiased evaluation and fair comparison. In supplementary materials section 3, we demonstrate an overview of our questionnaire design.

We assign three reviewers for each case, which comprises one original and three augmented reports. We meticulously balanced the workload among the seven reviewers and took measures to prevent case overlap in the questionnaires. For example, in the shuffled report assessments of Task 2, no clearly labeled versions from Task 1 are included, maintaining the integrity of the evaluation process. We conduct a thorough analysis of the returned questionnaires. In task 1, where all questionnaires are completed, an augmented report is considered acceptable if rated “Y”. The passing rates are high for all three styles: “*professional and concise*” at 99.2%, “*formal and concise*” at 98.5%, and “*formal and detailed*” at 98.1%.

For task 2, we received 2,351 valid assessments out of the potential 2,658 (886 original reports, each reviewed by three experts). Of these, 582 cases were reviewed thrice, 301 twice, and three once. As shown in Table 2, the original report ranks first in 41.2% of cases, while the “*formal and detailed*” style is most often ranked fourth. The ordinal Krippendorff’s Alpha of 0.573 indicates moderate agreement among reviewers, suggesting difficulty in reliably distinguishing between augmented and original data. Therefore, we exclude “*formal and detailed*” style reports from future model training for data quality assurance.

4.2.2 Main Results

Results on three datasets are shown in Table 1. “X-R2Gen” and “X-CvT2GPT2” denote our method *i.e.* adding the cVAE branch to the baseline models and training with augmented data. X-R2Gen-1 and X-R2Gen-2 represent the ratios of original to augmented data used in training, where X-R2Gen-1 corresponds to a 1:1 ratio and X-R2Gen-2 to a 1:2 ratio. Since we discard “*formal and detailed*” due to poor rating from expert evaluation, there is no X-R2Gen-3. The same naming convention applies to “X-CvT2GPT2”.

Table 3. Results on IU X-ray and MIMIC-mini using clinical efficacy metrics.

Dataset	Methods	Bert-S	F1	RE-ME	RE-NLI	Rad-C
IU X-ray	R2Gen[4]	0.539	0.404	0.410	0.449	0.280
	CvT2GPT2[21]	0.557	0.418	0.429	0.461	0.288
	X-R2Gen-2(Ours)	0.624	0.504	0.465	0.516	0.346
	X-CvT2GPT2-2(Ours)	0.645	0.523	0.488	0.535	0.366
MIMIC-mini	R2Gen[4]	0.489	0.401	0.330	0.204	0.182
	CvT2GPT2[21]	0.514	0.427	0.359	0.218	0.196
	X-R2Gen-2(Ours)	0.510	0.425	0.357	0.218	0.194
	X-CvT2GPT2-2(Ours)	0.521	0.436	0.368	0.222	0.201

After applying our method to the IU X-ray, the performance of baseline models is significantly improved and on par with SOTA models. The performance continues improving with increased training data volume. We apply several existing models with the released code base to MIMIC-mini and abdomen X-rays. The results are consistent with IU X-ray. As shown in Table 3, our method can also improve baseline models’ performance on CE metrics.

Bert-Score, CheXbert, and Radgraph-related metrics are based on models pre-trained on Chest X-rays or pre-extracted graphs on chest reports, which prohibits us from directly evaluating abdomen reports. To calculate CE metrics for abdomen reports, we first leverage the `en_core_sci_lg` model with UMLS linker from the `scispaCy` package to extract named entities and their corresponding labels from generated reports and ground truth reports, then calculate F1, Precision, and Recall based on the extracted labels. Results are shown in Table 4. Additionally, we conduct detailed analysis of the evaluation results of different style embeddings in Section 4 of the supplementary materials. We can observe that different style embeddings yield various results which proves the success of our decoupled training strategy. We also provide visualizations of generated reports in Section 2 of the supplementary materials.

Table 4. CE metrics of Abdomen X-ray.

Method	Micro			Macro		
	Precision	Recall	F1	Precision	Recall	F1
R2Gen	0.108	0.110	0.109	0.238	0.128	0.147
X-R2Gen-2	0.221	0.223	0.222	0.440	0.226	0.269
X-CvT2GPT2-2	0.276	0.269	0.272	0.419	0.250	0.285

4.3. Ablation Study

Data Volume & the Impact of cVAE Branch As illustrated in Table 5, data augmentation can indeed improve model performance. However, blindly increasing the amount of augmented data may impede model learning. This is primarily due to the distributional shift in writing styles induced by the augmented data. We suggest a solution by decoupling medical semantics from writing styles. Upon integrating a cVAE branch into the R2Gen, we observe a noteworthy enhancement in performance.

Different Techniques of Data Augmentation Based on the research of Chexbert [25], back-translation is proven to

Table 5. As the volume of augmented data increases, the performance of R2Gen first increases and then drops. A ratio of 1:0 signifies the absence of augmented data, a ratio of 1:1 indicates an equal amount of original and augmented data, and a ratio of 1:2 implies that the amount of the augmented data is twice that of the original data.

Dataset	Methods	Ratio	B-1	B-2	B-3	B-4	R-L	M	C
IU X-ray	w/o cVAE	1:0	0.470	0.304	0.219	0.165	0.371	0.187	-
		1:1	0.486	0.319	0.231	0.177	0.387	0.195	-
		1:2	0.430	0.248	0.162	0.162	0.329	0.168	-
	w/ cVAE	1:1	0.484	0.338	0.246	0.184	0.409	0.213	0.472
		1:2	0.522	0.369	0.271	0.206	0.406	0.218	0.488

Table 6. (Fr) indicates the model only uses original and French back-translation data. (Fr, De) indicates the model uses original data, as well as both French and German back-translations data.

Dataset	Methods	B-1	B-2	B-3	B-4	M	R-L	C
IU X-ray	X-R2Gen(Fr)	0.482	0.320	0.234	0.177	0.196	0.378	0.410
	X-R2Gen(Fr,De)	0.460	0.300	0.213	0.158	0.189	0.361	0.366
	X-R2Gen-2(Ours)	0.522	0.369	0.272	0.206	0.218	0.406	0.488
	X-CvT2GPT2(Fr)	0.488	0.327	0.242	0.181	0.207	0.389	0.485
	X-CvT2GPT2(Fr,De)	0.467	0.299	0.215	0.169	0.192	0.364	0.417
	X-CvT2GPT2-2(Ours)	0.525	0.370	0.275	0.209	0.240	0.409	0.655
MIMIC-mini	X-R2Gen(Fr)	0.372	0.229	0.150	0.103	0.140	0.282	0.139
	X-R2Gen(Fr,De)	0.361	0.216	0.139	0.094	0.137	0.270	0.126
	X-R2Gen-2(Ours)	0.380	0.233	0.155	0.109	0.143	0.283	0.156
	X-CvT2GPT2(Fr)	0.383	0.230	0.152	0.105	0.144	0.279	0.168
	X-CvT2GPT2(Fr,De)	0.364	0.216	0.141	0.102	0.137	0.271	0.144
	X-CvT2GPT2-2(Ours)	0.396	0.242	0.168	0.119	0.156	0.291	0.222

be an effective way of data augmentation *e.g.* they only experiment with English-German back-translation. To match the two different styles used for our data augmentation, we employ MarianNMT [15] to conduct two types of back-translations *i.e.* English-French and English-German, on the training set of IU X-ray and MIMIC-mini.

We rerun our models with the same hyperparameters on the back-translation augmented dataset. The results are shown in the Table 6. Data augmented by back-translation can still improve baseline model performance but falls far behind LLM-driven data augmentation.

5. Conclusion

This work introduces an agile and effective framework, X-Gen, for generating radiology reports across anatomical regions. We address the medical data scarcity challenge through LLM-driven data augmentation and fully utilize augmented data through decoupled training. We validate our approach on the IU X-ray dataset, MIMIC-mini, and an Abdomen X-ray dataset with two baseline models. The convincing results demonstrate the versatility and potential of our approach.

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